EDS 223 Homework Assignment 3

Prioritizing potential aquaculture

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Read in data

```
library(sf)          # vector handling
library(terra)       # raster handling
library(tidyverse)
library(tmap)        # map making
library(kableExtra) # table formatting
```

```
sst2008 <- rast('data/average_annual_sst_2008.tif')
sst2009 <- rast('data/average_annual_sst_2009.tif')
sst2010 <- rast('data/average_annual_sst_2010.tif')
sst2011 <- rast('data/average_annual_sst_2011.tif')
sst2012 <- rast('data/average_annual_sst_2012.tif')
depth <- rast('data/depth.tif')
regions <- st_read('data/wc_regions_clean.shp')
```

```
# stack sst
sst <- c(sst2008, sst2009, sst2010, sst2011, sst2012)
```

Process data

Find the mean SST from 2008-2012 (e.g. create single raster of average SST)

```
mean_sst_k <- app(sst, fun = 'mean')
summary(mean_sst_k)
```

```
      mean
 Min.   :282.1
 1st Qu.:285.6
 Median :287.2
 Mean   :287.3
 3rd Qu.:289.2
 Max.   :301.2
 NAs    :42347
```

Convert average SST from Kelvin to Celsius (hint: subtract by 273.15)

```
mean_sst <- mean_sst_k - 273
summary(mean_sst)
```

```
      mean
 Min.   : 9.082
 1st Qu.:12.552
 Median :14.241
 Mean   :14.323
 3rd Qu.:16.214
 Max.   :28.200
 NAs    :42347
```

Crop depth raster to match the extent of the SST raster

- note: the resolutions of the SST and depth data do not match

```
depth_crop <- crop(depth, ext(mean_sst))
depth_crop <- project(depth_crop, mean_sst)
```

Resample the depth data to match the resolution of the SST data using the bilinear method approach

```
depth_crop_resamp <- resample(depth_crop, mean_sst, method = 'bilinear')
```

Check that the depth and SST match in resolution, extent, and coordinate reference system

- hint: can the rasters be stacked?

```
# confirm extent matches
cat("Extent matches:", ext(depth_crop_resamp) == ext(mean_sst))
```

```
Extent matches: TRUE
```

```
cat("Resolution matches:", res(depth_crop_resamp) == res(mean_sst))
```

```
Resolution matches: TRUE TRUE
```

```
cat("CRS matches:", same.crs(depth_crop_resamp, mean_sst))
```

```
CRS matches: TRUE
```

Find suitable locations

```
# stack mean_sst and depth
mean_sst_depth <- c(mean_sst, depth_crop_resamp)
```

```
map_habitat <- function(mintemp, maxtemp, mindepth, maxdepth,name){
  # define reclassify for sst
  rcl_sst <- matrix(c(
    -Inf, mintemp, 0,
    mintemp, maxtemp, 1,
    maxtemp, Inf, 0
  ), ncol = 3, byrow = TRUE)

  # define reclassify for oyster depth
  rcl_depth <- matrix(c(
    -Inf, mindepth, 0,
    mindepth, maxdepth, 1,
    maxdepth, Inf, 0
  ), ncol = 3, byrow = TRUE)
```

```
  sst_zones <- as.factor(classify(mean_sst_depth[[1]], rcl = rcl_sst))
  depth_zones <- as.factor(classify(mean_sst_depth[[2]], rcl = rcl_depth))

  # Find suitable habitats
  habitat_zone = as.factor(sst_zones * depth_zones)

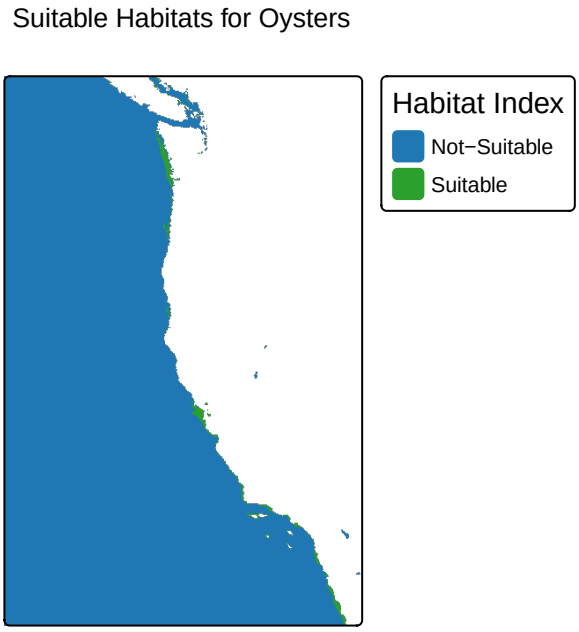
  levels(habitat_zone) <- data.frame(
    value = 0:1,
    label = c("Not-Suitable", "Suitable")
  )

  # Make map
  map <- tm_shape(habitat_zone) +
    tm_raster(col.legend = tm_legend("Habitat Index",
              position = tm_pos_out("right","center"))) +
    tm_title(paste("Suitable Habitats for",name))

  # Determine most suitable EEZ
  hab_area <- cellSize(habitat_zone, unit = 'km') |>
    mask(habitat_zone, maskvalues = 0)
  habitable_area <- zonal(hab_area, vect(regions), fun = 'sum', na.rm = TRUE)
  count_by_zone <- zonal(habitat_zone, vect(regions), fun = "sum", na.rm = TRUE)
  count_by_region <- cbind(region = regions$rgn,
                           count = count_by_zone,
                           area_km2 = habitable_area)
  colnames(count_by_region) <- c('Region', 'Count','Area(km^2)')

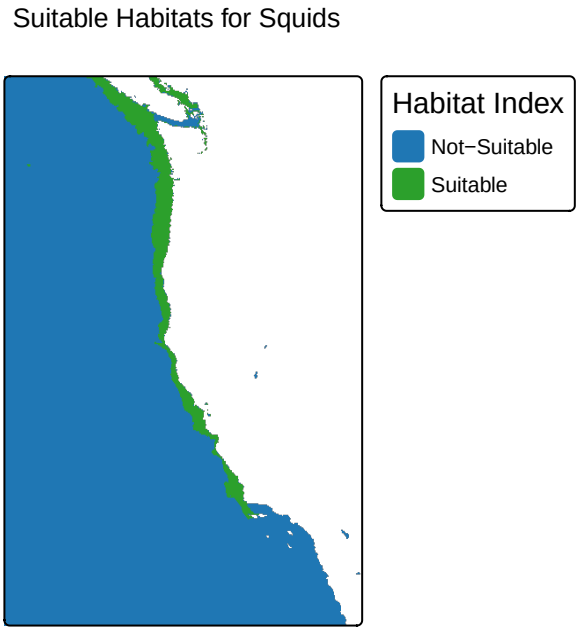
  # Display the outputs
  print(map)
  print(count_by_region)
}
```

```
# Make map of oysters
map_habitat(11,30,-70,0,"Oysters")
```



	Region	Count	Area(km^2)
1	Oregon	89	2547.540
2	Northern California	28	1140.895
3	Central California	218	4988.420
4	Southern California	172	3775.949
5	Washington	219	6226.850

```
# Make map of the common squid
map_habitat(10,14,-700,0,'Squids')
```



	Region	Count	Area(km^2)
1	Oregon	1650	26540.119
2	Northern California	746	12922.949
3	Central California	921	17123.111
4	Southern California	249	5119.809
5	Washington	1124	19358.250